

MANGALORE UNIVERSITY



State Education Policy – 2024
[SEP-2024]

CURRICULUM STRUCTURE

FOR

BSc in COMPUTER SCIENCE
(V and VI Semester)
(2026-27 Onwards)

MANGALORE UNIVERSITY

Suggested programme structure for the Under Graduate Programme

Bachelor of Science – B.Sc

Semester	Course1	Course2	Course3	Elective / Optional	Course	Language	Compulsory	Total Credit	Total Working hours
I	5 (3T+2P)	5 (3T+2P)	5 (3T+2P)			3+3	2	23	4+4+4+4+4+4+4+2=34
II	5 (3T+2P)	5 (3T+2P)	5 (3T+2P)			3+3	2	23	4+4+4+4+4+4+4+2=34
III	5 (3T+2P)	5 (3T+2P)	5 (3T+2P)	2		3+3		23	4+4+4+4+4+4+4+2=34
IV	5 (3T+2P)	5 (3T+2P)	5 (3T+2P)	2		3+3	2	25	4+4+4+4+4+4+4+2+2=36
V	8[(2x3T)+2P]	8[(2x3T)+2P]	8[(2x3T)+2P]				2	26	3+3+4+3+3+4+3+3+4+2=32
VI	8[(2x3T)+2P]	8[(2x3T)+2P]	8[(2x3T)+2P]				2	26	3+3+4+3+3+4+3+3+4+2=32
								146	202

Note:

- **Course1, Course2 and Course3: I to IV Semester:**
Theory 3 credits =4 contact hours & Practical 2 credits=4 contact hours
- **Course1, Course2 and Course3: V and VI Semester:**
Theory 3 credits=3 contact hours & Practical 2 credits =4 contact hours
- **Elective/Optional: 2 credits=2 contact hours**
- **Languages: 3 credits=4 contact hours**
- **Compulsory: 2 credits=2 contact hours**

BSc – COMPUTER SCIENCE

Semester V

Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BSCCSCS501	Programming in Python	Core	3	80	20	100	3
2	BSCCSCS502	Operating Systems	Core	3	80	20	100	3
3	BSCCSPS501	Python and Linux Lab	Practical	4	40	10	50	2
4	BSCCSSS503	Artificial Intelligence Tools	Compulsory	2	40	10	50	2

Semester VI

Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BSCCSCS601	Statistical Computing and R Programming	Core	3	80	20	100	3
2	BSCCSCS602	Web Technologies	Core	3	80	20	100	3
3	BSCCSPS601	R Programming and Web Technologies Lab	Practical	4	40	10	50	2
4	BSCCSSS603	Computer Hardware and Maintenance	Compulsory	2	40	10	50	2

Program Name	B.Sc	Semester	V
Course Title	Programming in Python (THEORY)		
Course Code:	BSCCSCS501	No. of Credits	3
Contact hours	39 Hours	Duration of SEA/Exam	3 hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Setup python to develop simple applications
- Understand the basic concepts in Python Programming
- Learn how to write, debug and execute Python programs
- Understand and demonstrate the use of advanced data types such as tuples, dictionaries and lists, Tuples and Sets
- Design solutions for problems using object-oriented concepts in Python
- Use and apply the different Python Libraries for GUI Interface, Data Analysis and Data Visualization.
- Extend the knowledge of python programming to build successful career in software development.

UNIT - I	9 Hours
Introduction to Features and Applications of Python; Python Versions; Installation of Python; Python Command Line mode and Python IDEs; Simple Python program. Python Basics: Identifiers; Keywords; Statements and Expressions; Variables; Operators; Precedence and Association; Data Types; Indentation; Comments; Built-in Functions: Console Input and Console Output, Type Conversions. Python Control Flow: Types of Control Flow; Control Flow Statements- if, else, elif, while loop, break, continue statements, for loop Statement; range () and exit () functions. Exception Handling: Types of Errors; Exceptions; Exception Handling using try, except and finally.	BOOK 2: Page No. 1 to 4, 10, 11, 31, 32 BOOK 1: Chapter 2 Complete BOOK 1: Chapter 3 (3.1 to 3.8)

Python Functions: Types of Functions; Function Definition-Syntax, Function Calling, Passing Parameters/arguments, the return statement; Default Parameters; Command line Arguments; Key Word Arguments; Recursive Functions; Scope and Lifetime of Variables in Functions.	BOOK 1: Chapter 4 Complete
UNIT - II	10 Hours
Strings: Creating and Storing Strings; Accessing String Characters; the str() function; Operations on Strings-Concatenation, Comparison, Slicing and Joining, Traversing; Format Specifiers; Escape Sequences; Raw and Unicode Strings; Python String Methods. Lists: Creating Lists; Operations on Lists; Built-in Functions on Lists; Implementation of Stacks and Queues using Lists; Nested Lists. Dictionaries: Creating Dictionaries; Operations on Dictionaries; Built-in Functions on Dictionaries; Dictionary Methods; Populating and Traversing Dictionaries. Tuples and Sets: Creating Tuples; Operations on Tuples; Built-in Functions on Tuples; Tuple Methods; Creating Sets; Operations on Sets; Built-in Functions on Sets; Set Methods.	BOOK 1: Chapter 5 (5.1 to 5.5) BOOK 1: Chapter 6 (6.1 to 6.5) BOOK 1: Chapter 7 (7.1 to 7.4) BOOK 1: Chapter 8 (8.1 to 8.4, 8.7, 8.9, 8.10)
UNIT – III	10 Hours
File Handling: File Types; Operations on Files– Create, Open, Read, Write, Close Files; File Names and Paths; Format Operator. Object Oriented Programming: Classes and Objects; Creating Classes and Objects; Constructor Method; Classes with Multiple Objects; Objects as Arguments; Objects as Return Values; Inheritance- Single and Multiple Inheritance, Multilevel and Multipath Inheritance; Encapsulation-Definition, Private Instance Variables; Polymorphism-Definition, Operator Overloading.	BOOK 1: Chapter 9 (9.1 to 9.3) BOOK 1: Chapter 11 (11.1 to 11.5, 11.7 to 11.9)
UNIT - IV	10 Hours
Python SQLite: The SQLite3 module; SQLite Methods-connect, cursor, execute, close; Connect to Database; Create Table; Operations on Tables Insert, Select, Update. Delete and Drop Records. Data Analysis: NumPy- Introduction to NumPy, Array Creation using NumPy, Operations on Arrays; Pandas-Introduction to Pandas, Series and DataFrames, Creating DataFrames from Excel Sheet and .csv file, Dictionary and Tuples. Operations on DataFrames.	Study Material BOOK 1: Chapter 12 12.3 to 12.3.5, 12.4, to 12.4.1 (Upto Page Number 379 only) BOOK 2: Page No: 694 to 703

Data Visualisation: Introduction to Data Visualisation; Matplotlib Library; Different Types of Charts using Pyplot- Line chart, Bar chart and Histogram and Pie chart.	BOOK 2: Page No: 705 to 712
Text Books: 1. Introduction to Python Programming by Gowrishankar S. and Veena A., CRC Press. 2. Core Python Programming by R. Nageswara Rao, Dreamtech. Reference Books: 1. Think Python How to Think Like a Computer Scientist, Allen Downey et al., 2nd Edition, Green Tea Press. Freely available online @ https://www.greenteapress.com/thinkpython/thinkCSpy.pdf, 2015. 2. Introduction to Python Programming, Gowrishankar S et al., CRC Press, 2019. 3. Python Data Analytics: Data Analysis and Science Using Pandas, matplotlib, and the Python Programming Language, Fabio Nelli, Apress®, 2015 4. Advance Core Python Programming, MeenuKohli, BPB Publications, 2021. 5. Core PYTHON Applications Programming, Wesley J. Chun, 3rd Edition, Prentice Hall, 2012. 6. Automate the Boring Stuff, Al Sweigart, No Starch Press, Inc, 2015. 7. Data Structures and Program Design Using Python, D Malhotra et al., Mercury Learning and Information LLC, 2021.	

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning.

Program Name	B.Sc	Semester	V
Course Title	Operating System (THEORY)		
Course Code:	BSCCSCS502	No. of Credits	3
Contact hours	39 Hours	Duration of SEA/Exam	3 hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

At the end of the course, students will be able to:

- Understand the fundamentals of the operating system.
- Comprehend multithreaded programming, process management, process synchronization, memory management and storage management.
- Compare the performance of Scheduling Algorithms.
- Identify the features of I/O and File handling methods.
- Linux commands and shell programming.

UNIT - I	9 Hours
Introduction: Operating System, Simple Batch Systems, Multi programmed Batched Systems, Time Sharing Systems, Real-Time Systems, Multi-processor Systems. System Components, Operating System Services. Process: Process Concept, Process Scheduling, Cooperating Process, CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms.	BOOK 1 Chapter 1: 1.1, 1.1.1, 1.1.2, 1.2, 1.2.1, 1.2.2, 1.2.3, 1.4, 1.7. Chapter 3: 3 3.1, 3.1.1 to 3.1.8, 3.2. Chapter 4: 4.1, 4.1.1 to 4.1.4, 4.2, 4.2.1 to 4.2.3, 4.4. Chapter 6: 6.1, 6.2 6.3
UNIT - II	12 Hours
Process Synchronization: Introduction Race Condition, Critical Section problem and Peterson's Solution. Semaphores: Classic problems of synchronization, Readers-writers problem, Dining Philosophers problem. Deadlocks: Deadlock Characterization, Methods of Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	Chapter 6: 7.1, 7.2, 7.4 , 7.5 Chapter 8: 8.2, 8.2.1, 8.2.2, 8.3, 8.4, 8.4.1 to 8.4.4, 8.5, 8.5.1 to 8.5.3 (8.5.3.1 and 8.5.3.3), 8.6, 8.6.1 – 8.6.2, 8.7, 8.7.1, 8.7.2.

UNIT – III	8 Hours
Memory Management. Logical versus Physical Address Space, Swapping, Contiguous Allocation (Memory Allocation, Fragmentation), Paging (Basic Method), Segmentation (Basic Method). Virtual Memory. Demand Paging, Page Replacement, Page Replacement Algorithms. File System: File Concept and access methods	Chapter 9: 9.1.2, 9.2, 9.3, 9.3.2, 9.3.3, 9.4, 9.4.1, 9.5, 9.5.1. Chapter 10: 10.1, 10.2, 10.2.1, 10.4.1 to 10.4.4, 10.6. Chapter 11: 11.1, 11.1.1, 11.1.2, 11.2
UNIT - IV	10 Hours
Linux files system, login and logout. Linux commands: Directory oriented command, wild card characters, File oriented commands, File Access Permissions, Process oriented commands, Background processing, Communication oriented commands, General purpose commands. Pipe and Filter related commands, vi editor, Shell programming.	BOOK 2 Chapter 2: Page 8-33, Page 36-37 (Exclude du and df, ln, comm, touch, expand, nl, tac, tail, head, nohup, at, batch), Pipe and Filters related commands. vi Editor, Shell programming, System administration. Chapter 3: Page 39-52. (Exclude egrep, fgrep, uniq, pr, sed, gawk & also exclude pages 56-65). Chapter 4: Page 66-71. Chapter 5: Page 72-76 & Page 80-92. (Exclude command export & exclude base name). Chapter 7: Page 119- 123 (Exclude Managing Devices)
Text Books: 1. Abraham Silberschartz, Peter B Galvin, Greg Gange, Operating System Concepts, 6th edition, TMH 2. B. Mohamed Ibrahim, Linux: A Practical Approach, Firewall media, 2009 Reference Books: 1. Richard Peterson, Linux: The Complete Reference, 6th edition, Tata McGraw Hill Publisher.	

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning.

Program Name	BSc	Semester	V
Course Title	Artificial Intelligence Tools (Skill Based - Compulsory)		
Course Code:	BSCCSES503	No. of Credits	02
Contact hours	26 Hours	Duration of SEA/Exam	2 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Course Outcomes (COs):

At the end of the course, students will be able to:

- Identify and describe different types of intelligent agents and their environments, and implement basic agent-based systems.
- Use the some of the current AI tools.

Q.No	Works to be carried by the students
1	Create an invitation (image format) for the College Annual Day Programme by giving appropriate prompts to AI Tools such as Gemini, ChatGPT, Adobe Firefly or Canva
2	Create a thirty seconds video story about a historical or mythological event using any video creation AI Tool.
3	Create a Presentation about any topic using Gamma or NoteBookLM or any other Presentation creation AI Tool.
4	Create a Shopping Cart web Application using Claude, Caffeine, Gemini or Antigravity or any other Web Development AI Tool.
5	Create a dynamic web application which contains a mock test or a Quiz.

Evaluation:

The report shall be prepared by the student under the guidance of the identified mentor in the college and submitted to the Head of the Department for evaluation. The report shall be evaluated by the two internal faculty members and submit the summative marks to the university.

Scheme of Evaluation

Internal Assessment	10 Marks (C1) Basis for Internal Assessment Continuous Evaluation: 10 Marks
Evaluation of Report	40 Marks (C2) i) Preparation and presentation: 20 Marks ii) Viva-Voce: 10 Marks iii) Record Book: 10 Marks

Program Name	BSc	Semester	V
Course Title	Python Programming and Linux Lab - Practical		
Course Code:	BSCCSPS501	No. of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Python Programming Lab

PART – A

1. Write a program create list with N elements. find all unique elements in the list. If an element is found only once in the list, then add that element to the unique list.
2. Write a program, using user-defined functions to find the area of rectangle, square, circle and triangle by accepting suitable input parameters from user.
3. Write a program using user defined exception class that will ask the user to enter a number until he guesses a stored number correctly. To help them figure it out, a hint is provided whether their guess is greater than or less than the stored number using user defined exceptions.
4. Write a program to create a text file and compute the number of characters, words and lines in a file.

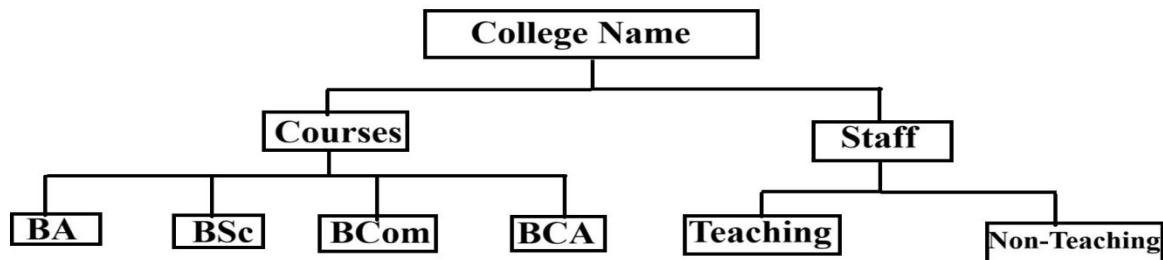
PART – B

1. Write a Pandas program to join the two given data frames along rows. Sample Data frame may contain details of student like rollno , name , Total Marks.
2. Program to create a class Employee with empno, name, depname, designation, age and salary and perform the following function.
 - a. Accept details of N employees
 - b. Search given employee using empno
 - c. Display employee details in neat format.
3. Create a table student table (regno, name and marks in 3 subjects) using MySQL/ SQLite and perform the followings
 - a. To accept the details of students and store it in database.
 - b. To display the details of all the students
 - c. Delete particular student record using regno.

Linux Lab

PART-A

1. Create the following directory structure:



- Put a file under BA, BSc, BCom, BCA, Teaching and Non-Teaching directory with meaningful content.
 - Goto BA directory and then move its file to Non-Teaching directory.
 - Change the directory to Non-Teaching from root directory using a single command.
 - From BA copy the content to B.Com. Delete BA directory recursively
 - List the files in long format. Starting from root directory including sub directory contents
 - Count how many files exist in office directory.
 - Display the present path.
 - Display current date in the form MM/DD/YY HH:MM:SS
2. Create a file which contains students' information such as Rollno, Name, Gender, class (BCA, BSc, B.com), Total marks.
- Append the 2 more records.
 - Display the contents.
 - Display the records of only BSc course and sort on reverse order of name and store in BSCnew.txt.
 - Display the contents of BSCnew.txt
 - Rename the student.dat to stud.dat
- Add the following permission
- Execute for user.
 - Remove write permission for group
 - Check the changes in the permission by listing the file.
- Select only the female names and store in female.dat and male names to male.dat
 - Combine these names from both the files. Show the new file.
 - List all the files that start with "B" in the current directory.

3. Write a shell script to accept 'n' and find the sum of the series 1!+3!+5!.....+n!.
4. Write a shell script to accept many characters and count individual vowels, digits, spaces, special characters and consonants.

PART-B

5. Write a shell script to accept a word and check whether it begins with lowercase vowel or uppercase vowel, ends with a digit or whether it is a three-letter word.
6. Write a menu driven shell script for the following:
 - i) Display the current working directory.
 - ii) Rename a file (check for the existence of the source file)
 - iii) List the users logged in.
 - iv) Append the contents of a file to another file (display the message if the file doesn't exist in the directory).
7. Write a menu driven shell script for the following.
 - a. Rename a file (check for the existence of the source file)
 - b. List all file names/ directory names in the present working directory which has the specified pattern
 - c. List of directories having all the permission
 - d. List only files names in long format.

Evaluation Scheme for Lab Examination

Assessment Criteria		
Question-1	One question from PART-A of Python or Linux Writing: 5 Marks Logic : 5 Marks Execution: 5 Marks	15 Marks
Question-2	One question from PART-B of Python or Linux Writing: 7 Marks Logic : 7 Marks Execution: 6 Marks	20 Marks
Practical Record		5 Marks
Total		40 Marks

Note:

- If the Question-1 is from Python, then the Question-2 must be from Linux.
- If the Question-1 is from Linux, then the Question-2 must be from Python.

Semester VI

Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BSCCSCS601	Statistical Computing and R Programming	Core	3	80	20	100	3
2	BSCCSCS602	Web Technologies	Core	3	80	20	100	3
3	BSCCSPS601	R Programming and Web Technologies Lab	Practical	4	40	10	50	2
4	BSCCSSS603	Computer Hardware and Maintenance	Compulsory	2	40	10	50	2

Program Name	B.Sc	Semester	VI
Course Title	Statistical Computing and R Programming		
Course Code:	BSCCSCS601	No. of Credits	3
Contact hours	39 Hours	Duration of SEA/Exam	3 hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Explore fundamentals of statistical analysis in R environment.
- Describe key terminologies, concepts and techniques employed in Statistical Analysis.
- Define Calculate, Implement Probability and Probability Distributions to solve a wide variety of problems.
- Conduct and interpret a variety of Hypothesis Tests to aid Decision Making.
- Understand, Analyse, and Interpret Correlation Probability and Regression to analyse the underlying relationships between different variables.

UNIT - I	9 Hours
Introduction of the language, numeric, arithmetic, assignment, and vectors, Matrices and Arrays, Non-numeric Values, Lists and Data Frames.	BOOK 1: Chapter 1: 4-12 Chapter 2: 17-36 Chapter 3: 39-58 Chapter 4: 59-85 Chapter 5: 89-101
UNIT -II	10 Hours
Reading and writing files, Programming: Calling Functions, Conditions and Loops, stand- alone statement with illustrations in exercise, stacking statements, coding loops, Writing Functions, Exceptions. Basic Data Visualization.	BOOK 1: Chapter 8: 147-161 Chapter 9: 165-176 Chapter 10:179-212 Chapter 11: 215-238 Chapter 12:241-258 Chapter 14:289-306

UNIT -III	10 Hours
Descriptive Statistics: Types of Data, Nominal, Ordinal, Scale and Ratio, Measures of Central Tendency, Mean, Mode and Median, Percentiles, Quartiles, Measures of Variability, Mean Absolute Deviation, Range, Inter-Quartile-Range, Standard Deviation, Z-Scores. Coefficient of Variation, Measure of shapes-Skewness and Kurtosis. Charts: Bar Chart, Pie Chart, Frequency Polygon, Stem and Leaf Diagram. Distributions: Binomial, Poisson, Continuous Probability Distribution, Normal Distribution, Uniform Distribution. Estimating the population mean using the t-distribution.	BOOK 2: 1.4(Page No 8-11) 3.1,3.2,3.3,3.4(Page No 53-84). 2.2(Frequency Polygon -Page No-25, Stem and Leaf -Page No -27). 2.3(Bar Chart, Pie Chart, Page No- 31and 32), Stem and Leaf Plots-Page No-27). 5.1,5.2,5.3,5.4(Page No 144-164). 6.1,6.2:-(Page No 185-199). 8.2:-(Page No 266-270).
UNIT -IV	10 Hours
Statistical Inference and Hypothesis Testing: Types of Hypotheses, and Sample, Null and Alternate Hypothesis, Level of Significance, Type I and Type II Errors, One Sample t-test, Chi Square Test of Independence, One way ANOVA. Correlation and Regression: Analysis of Relationship, Positive and Negative Correlation, Perfect Correlation, Karl Pearson Coefficient of Correlation, Correlation Matrix, Scatter Plots, Simple Regression Analysis.	BOOK 2: 9.1(Page No 296-301, 304-305) 9.3(Page No 314-318) 16.1(Page No 660-668) 11.2(Page No 412-420) 12.1(Page No 472-476)
Text Books: 1. Tilman M. Davies, “The book of R: A first course in programming and statistics”, San Francisco, 2016. 2. Ken Black, Business Statistics, New Delhi, Wiley, 2013. References: 1. Vishwas R. Pawgi, “Statistical computing using R software”, Nirali prakashan publisher, e1 edition, 2022.	

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning.

Program Name	BSc	Semester	VI
Course Title	Web Technologies (THEORY)		
Course Code:	BSCCSCS602	No. of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After completing this course satisfactorily, a student will be able to:

- Understand the basics of PHP and MySQL, including their roles in web development.
- Write and execute PHP scripts to create dynamic web pages.
- Use PHP sessions and cookies to manage state and user information.
- Understand the fundamentals of JavaScript, including its syntax and data types.
- Write and execute JavaScript code to create interactive web pages.
- Implement event handling to create responsive user interfaces.

UNIT - I	9 HOURS
Basics of HTML and CSS3. Animation: Introduction, Start and End States, Interpolation, Animations in HTML. All About CSS Animations, creating a Simple Animation, Detailed Look at the CSS Animation Property, Keyframes, Declaring Multiple Animations, Wrap-up. All About CSS Transitions, Adding a Transition.	Book 1: Page no. (9 – 49)
UNIT - II	10 HOURS
Introduction to PHP: Introduction to PHP, History and Features of PHP, Understanding PHP, Writing Comments in PHP, Sending Data to the Web Browser. Data types in PHP, Keywords in PHP. Using Variables, Constants in PHP, Expressions in PHP, Operators in PHP, Conditional statements: if, if-else, switch, The ? Operator, Looping statements: while Loop, do while Loop, for Loop.	Book 4: Chapter 1(1 – 13) Book 2: Chapter 2(21 – 29) Book 3: Chapter 3(65 – 72, 83,84), Chapter 4 (95 - 123)
UNIT - III	10 HOURS
Arrays in PHP: Creating Arrays, Accessing Array elements, Multidimensional arrays. Indexed v/s Associative arrays, using Array Functions. Functions in PHP: Function definition, Creating and invoking user-defined functions, Formal parameters versus actual parameters, Function and variable scope.	Book 4: Chapter 2 (54 - 61) Book 2: Chapter 5 (119) Book 3: Chapter 6 (173 - 179)

Strings in PHP: What is String? Creating and Declaring String, String Functions. Class and Objects in PHP: Creating and accessing Class & Object, Object properties, object methods.	Book 2: Chapter 3 (64-74) Chapter 4 (77-86, 92- 96) Chapter 6 (148 - 155)
UNIT - IV	10 HOURS
Form Handling: Creating HTML Form, Handling HTML Form data in PHP Database Handling using PHP with MySQL: Introduction to MySQL, Data Types, Accessing MySQL using MySQL Client and Using phpMyAdmin, MySQL Commands, Using PHP with MySQL, PHP MySQL Functions, connecting to MySQL and Selecting the Database, Executing Simple Queries, Retrieving Query Results, Counting Returned Records, Inserting, deleting and Updating Records with PHP.	Book 3: Chapter 12 (340 - 365) Book 4: Chapter 4 (111 -127) Chapter 5 (130 - 151) Chapter 9 (265 - 292)

Textbooks:

1. Animation in HTML, CSS, and JavaScript, Kirupa Chinnathambi, 1st Edition, CreateSpace Independent Pub, 2013.
2. Kevin Tatroe, Peter MacIntyre & Rasmus Lerdorf, Programming PHP, 3rd Edition, O'Reilly
3. Robin Nixon, Learning PHP, MySQL, JavaScript, CSS & HTML5, 3rd Edition, O'Reilly (Soft Copy)
4. Larry Ullman, PHP and MySQL for Dynamic Web Sites, 4th Edition, Peachpit Press

Reference Books:

1. Deitel, Paul_Deitel, Harvey_Deitel, Abbey - Internet and World Wide Web How to Program-Pearson Education (US) (2011)
2. Steven Holzner, PHP the Complete Reference
3. Chris Bates - Web Programming_ Building Internet Applications-Wiley (2002)
4. Web Programming with HTML5, CSS, and JavaScript-John Dean.

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BSc	Semester	VI
Course Title	R Programming and Web Technologies Lab - Practical		
Course Code:	BSCCSES603	No. of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

R Programming Lab

PART-A

1. Write an R program to create a list containing strings, numbers, vectors and logical values and do the following manipulations over the list
 - a. Access the first element in the list
 - b. Give the names to the elements in the list
 - c. Add element at some positions in the list
 - d. Remove the element
 - e. print the first and third element
 - f. Update the third element

2. The following table shows the time taken (in minutes) by 100 students to travel to school on a particular day.

Time	0-5	5-10	10-15	15-20	20-25
No. of students	5	25	40	17	13

- a. Draw the histogram
 - b. Draw frequency polygon

3. Create a factor marital_status with levels Married, single, divorced. Perform the following operations on this factor
 - a. Check the variable is a factor
 - b. Access the 2nd and 4th element in the factor
 - c. Remove third element from the factor
 - d. Modify the second element of the factor
 - e. Add new level widowed to the factor and add the same level to the factor marital_status

4. Write an R program to create a Data Frame with following details and do the following operations.

ItemCode	itemCategory	ItemPrice
1001	Electronics	700
1002	Desktop Supplies	300
1003	Office Supplies	350
1004	USB	400
1005	CD Drive	800

- Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.
- Subset the Data frame and display only the items where the category is either “Office Supplies” or “Desktop Supplies”.
- Subset the Data frame and display the items where the Itemprice between 300 and 700
- Compute the sum of all ItemPrice.
- Create another Data Frame called “item-details” with three different fields itemCode, ItemQtyonHand and ItemReorderLvl and merge the two frames.

PART-B

1. Write a R program to create a Vector containing following 8 values

4 3 0 5 2 9 4 5

and perform the following operations:

- Find mean, median, mode.
 - Find the range.
 - Find the 35th and 78th percentile.
 - Find the variance and standard deviation
 - Find the interquartile range.
 - Find the z-score for each value.
2. Write R script to compute the regression equation of y on x from the following data. Predict the value of y when x=7.

X	2	4	5	6	8	11
Y	18	12	10	8	7	5

3. Write R script to find the correlation coefficient and type of correlation between advertisement expenses and sales volume using Karl Pearson's coefficient of correlation method (Direct Method).

Firm	1	2	3	4	5	6	7	8	9	10
Advertisement Exp. (Rs. In Lakhs)	11	13	14	16	16	15	15	14	13	13
Sales Volume (Rs. In Lakhs)	50	50	55	60	65	65	65	60	60	50

Web Technologies Lab

PART-A

1. Write a HTML5 program to develop a static Web Page for Catalog. Also place a SUBMIT button on this page.

	Book:XML Bible Author:Wingston Publication:wuely	\$40.5	add to cart
	Book:A1 Author:S.Rusell Publication:Princeton hall	\$63	add to cart
	Book:JAVA 2 Author:Watson Publication:BPB	\$35.5	add to cart
	Book:Html in 24 hrs Author:Sam Peter Publication:SAM	\$50	add to cart

For each click on “ Add Cart” of any book. On clicking Submit button show total amount to be paid in an alert box.

2. Write a PHP program with a user interface for calculating the number of days between two given dates. Display the difference in years, months, and days.

Start Date

May ▾

13 ▾

2026



End Date

Jul ▾

21 ▾

2027



☐ include end day (add 1 day)

Calculate



3. Design and implement a web page using HTML and CSS to create a box that performs multiple animations simultaneously.
4. Write a PHP program that includes a user form with a text field and submit buttons for various string manipulations. It will display the result accordingly. (for replace, replace 'a' with 'x').

String Manipulation

Enter a string:

PART-B

1. Create a PHP webpage for an **Online Product Gallery** that displays product cards with images, product name, price, and description fetched using PHP variables or arrays.
Apply **CSS transitions and transformations** so that when the user hovers over a product card:
 - a) the card smoothly enlarges
 - b) rotates slightly
 - c) changes its shadow and opacity and
 - d) the image zooms in with different timing functions
2. Write a PHP program that implements a class to add and find the difference of two distance values given in feet and inches. The user inputs are collected through an HTML form.

Distance Calculator

Distance 1:
Feet:
Inches:

Distance 2:
Feet:
Inches:

Results:

Sum: 9' 5"

Difference: 5' 8"

3. Create a PHP-MySQL application for a Student Attendance Management System. Create a MySQL database and a table to store student attendance details such as Student USN, Student Name, Class, Subject, Attendance Percentage, Attendance Status. Design a webpage using HTML and CSS to input the above fields. Apply appropriate CSS styling and validation for input fields. Use PHP to establish a connection with the MySQL database, insert student attendance details into the database table, fetch and display all records in a formatted table. Provide options to Save Record, Clear Form and View Attendance Details.

Evaluation Scheme for Lab Examination

Assessment Criteria		
Question-1	One question from PART-A of R Programming or Web Technologies Writing: 5 Marks Logic : 5 Marks Execution: 5 Marks	15 Marks
Question-2	One question from PART-B of R Programming or Web Technologies Writing: 7 Marks Logic : 7 Marks Execution: 6 Marks	20 Marks
Practical Record		5 Marks
Total		40 Marks

Note:

- If the Question-1 is from R Programming, then the Question-2 must be from Web Technologies.
- If the Question-1 is from Web Technologies, then the Question-2 must be from R Programming.

Program Name	BSc	Semester	VI
Course Title	Computer Hardware and Maintenance (Skill Based - Compulsory)		
Course Code:	BSCCES 603	No. of Credits	02
Contact hours	26 Hours	Duration of SEA/Exam	2 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Course Objectives:

After completing this course satisfactorily, a student will be able to understand:

- Microsoft Windows Administration
- Maintenance

Q.No	Activities to be carried by the students
1	Exercise on windows settings and control panel tools
2	Exercise on user management and disk management
3	Exercise on Windows firewall and command line tools
4	Exercise on Windows networking
5	Exercise on Windows administrative tools.

Evaluation:

The report shall be prepared by the student under the guidance of the identified mentor in the college and submitted to the Head of the Department for evaluation. The report shall be evaluated by the two internal faculty members and submit the summative marks to the university.

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning.

Scheme of Evaluation

Internal Assessment	10 Marks (C1) Basis for Internal Assessment Continuous Evaluation: 10 Marks
Evaluation of Report	40 Marks (C2) iv) Preparation and presentation: 20 Marks v) Viva-Voce: 10 Marks vi) Record Book: 10 Marks

Question Paper Pattern for Core Subjects

Duration:3 Hours

Max. Marks: 80

Note: Answer any ten Questions from Part-A and one full Question from each unit of Part-B

PART-A

1. Answer any 10

10 x 2=20

- a.
- b.
- c.
- d.
- e.
- f.
- g.
- h.
- i.
- j.
- k.
- l.

PART-B

There are 4 Units - UNIT-I, II, III, IV

Each unit contains 2 main questions and it carry 15 marks.

4 x 15=60

Each main question contains 2 or more sub questions.

UNIT-I

2 a)

b)

c)

3. a)

b)

c)

Same pattern for UNIT – II , UNIT - III and UNIT - IV